COINCIDENCE OF ABSOLUTE AND RELATIVE PSEUDO-COHERENCE FOR FLAT FINITE PRESENTATION QUOTIENTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Flat and finite presentation ring maps are perfect, hence pseudo-coherent ring maps. invoked at proof lines 21901-21903. Makes relative pseudo-coherence over R moot.
- Over pseudo-coherent ring maps absolute and relative pseudo-coherence agree. invoked at proof lines 21904-21905. Gives (1) iff (2) and (4) iff (5).
- Relative pseudo-coherence is transitive under composition, and A is pseudo-coherent relative to R. invoked at proof lines 21908-21910. Gives (2) iff (3).
- The definition of relative pseudo-coherence over a surjective ring map reduces to absolute pseudo-coherence of the restriction. used at proof lines 21911-21913. Gives (3) iff (4).

Source reference. Original lemma 067J: sources/stacks-more-algebra/pinned-original-source.tex, source lines 21815–21815. Cited at proof lines 21901-21902 to identify the ring maps as perfect; the equivalence assembly is inside the counted span.

Source reference. Original lemma 0DHQ: sources/stacks-more-algebra/pinned-original-source.tex, source lines 21870–21870. Cited at proof lines 21904-21905 for (1) iff (2) and (4) iff (5); the remaining equivalences are inside the counted span.

Source reference. Original lemma 067D: sources/stacks-more-algebra/pinned-original-source.tex, source lines 21638–21638. Cited at proof lines 21908-21910 for (2) iff (3); the surjective-case equivalence is inside the counted span.

Source reference. Original definition 065I: sources/stacks-more-algebra/pinned-original-source.tex, source lines 21332–21332. Cited at proof lines 21912-21912 for the direct equivalence (3) iff (4) under the surjection; the assembly is inside the counted span.

Lemma 0.2. *Let $R \to B \to A$ be ring maps with $\varphi : B \to A$ surjective and $R \to B$ and $R \to A$ flat and of finite presentation. For $K \in D(A)$ denote $\varphi_* K \in D(B)$ the restriction. The following are equivalent*

(1) *K is pseudo-coherent,*
(2) *K is pseudo-coherent relative to R,*
(3) *K is pseudo-coherent relative to A,*
(4) *$\varphi_* K$ is pseudo-coherent,*

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(5) $\varphi_* K$ *is pseudo-coherent relative to* R.

Similar holds for m*-pseudo-coherence.*

Proof. Observe that $R \to A$ and $R \to B$ are perfect ring maps (Lemma 067J) hence a fortiori pseudo-coherent ring maps. Thus (1) $\Leftrightarrow$ (2) and (4) $\Leftrightarrow$ (5) by Lemma 0DHQ.

Using that A is pseudo-coherent relative to R we use Lemma 067D to see that (2) $\Leftrightarrow$ (3). However, since $A \to B$ is surjective, we see directly from Definition 065I that (3) is equivalent with (4). $\square$